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Metabolites identification and multi-component pharmacokinetics of ergostane and lanostane triterpenoids in the anticancer mushroom *Antrodia cinnamomea*

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Abstract

Antrodia cinnamomea is a precious medicinal mushroom popularly used for adjuvant cancer therapy in Taiwan. Its major bioactive constituents are ergostane and lanostane triterpenoids. Although clinical trials for *A. cinnamomea* have been recently initiated, its metabolism remains unclear. The present study aims to elucidate the metabolism and pharmacokinetics of *A. cinnamomea* in rats. After oral administration of an ethanol extract, 18 triterpenoids and 8 biotransformed metabolites were detected in rats plasma by UHPLC/qTOF-MS. Four of the metabolites were prepared by semi-synthesis and fully identified by NMR, while the others were tentatively characterized by comparing with the metabolites of single compounds (antcins B, C, H and K). Furthermore, a multi-component pharmacokinetic study of *A. cinnamomea* was carried out to monitor the plasma concentrations of 14 triterpenoids (ergostanes **1**–**3**, **5**–**8**, **14**–**16**; lanostanes **9**, **10**, **17**, **19**) and 2 metabolites (**M5**, **M6**) by LC/MS/MS in rats after oral administration of the ethanol extract (1.0 g/kg). The results showed that ergostanes and $\Delta^{7,9(11)}$ lanostanes, but not Δ^{8} lanostanes, could get into circulation. The low-polarity ergostanes (antcins B and C) undertook hydrogenation (C-3 or C-7 carbonyl groups) or hydroxylation to produce polar metabolites. High-polarity ergostanes (antcins H and K) and $\Delta^{7,9(11)}$ lanostanes were metabolically stable. We also discovered that ergostanes and lanostanes showed remarkably different pharmacokinetic patterns. The ergostanes were generally absorbed and eliminated rapidly, whereas the lanostanes remained in the plasma at a low concentration for a relatively long time. The results indicate that high-polarity ergostanes are the major

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plasma-exposed components of *A. cinnamomea*, and may play an important role in its therapeutic effects.

Keywords

Antrodia cinnamomea; ergostane; lanostane; metabolites identification; multi-component pharmacokinetics; triterpenoids.

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1. Introduction

Antrodia cinnamomea (*Antrodia camphorata*, Polyporaceae family) is a precious medicinal mushroom [1]. Known as the traditional medicine Niu-Chang-Chih, it has been popularly used to treat cancer, intoxication and inflammation for a long history in Taiwan [2,3]. According to a recent survey, 12% cancer patients in Taiwan use *A. cinnamomea* as adjuvant therapeutic agent or nutrient during cancer treatment [4]. Preclinical studies have revealed that *A. cinnamomea* could inhibit tumor growth by 42% and 80% in mice bearing H22 liver tumor and MDA-MB-231 breast tumor xenograft, respectively [5,6]. *A. cinnamomea* is also widely used by healthy people as a dietary supplement to promote physical strength. Clinical trials for *A. cinnamomea* have been initiated recently [7,8]. However, little is known on *in vivo* metabolism and pharmacokinetics of this multi-component traditional medicine, so far.

A. cinnamomea contains abundant ergostane and lanostane tetracyclic triterpenoids, accounting for 30% of its methanol extract [3]. Ergostanes (C-4 α-CH_3, $\Delta^{24,28}$) are its characteristic constituents. Although lanostanes are widely distributed in medicinal fungi such as *Ganoderma lucidum* (Ling-Zhi) and *Poria cocos* (Fu-Ling), most lanostanes from *A. cinnamomea* have lower degree of oxygenation (1-3 –OH/=O groups) [9]. Many of these triterpenoids show anti-cancer and anti-inflammatory activities [10,11]. For example, dehydroeburicoic acid (**19**, 10 μg/g dosage) could significantly inhibit tumor growth (0.16 g *vs* 0.29 g, $p < 0.05$) in HL 60 cell xenograft mice; antcin C (**5**/**6**) could protect 2,2-azobis(2-amidinopropane)

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dihydrochloride-induced mice liver damage via Nrf2 pathway [12,13]. Thus, triterpenoids are generally considered as the major bioactive constituents of *A. cinnamomea*, and are used as chemical markers for its quality control [14]. Metabolic studies of *Ganoderma lucidum* and *Poria cocos* showed that lanostane triterpenoids were orally bioavailable, and mainly underwent hydrogenation, hydroxylation, and dehydrogenation metabolic reactions *in vivo* [15,16]. However, the *in vivo* metabolism of ergostanes and lanostanes in *A. cinnamomea* has never been reported. Furthermore, most triterpenoids in *A. cinnamomea* have poor solubility in water and even ethanol or acetone. Their absorption after oral administration, as well as their *in vivo* metabolic pathway and plasma concentrations warrants to be clarified.

The present work studied the metabolism and pharmacokinetics of *A. cinnamomea* in rats. After oral administration of an ethanol extract, 18 triterpenoids and 8 metabolites were detected in rats plasma by ultra-high performance liquid chromatography coupled with quadrupole time-of-flight mass spectrometry (UHPLC/qTOF-MS). The metabolites were identified by comparing with reference standards, or by comparing to metabolites of single compounds (antcins B, C, H and K). A multi-component pharmacokinetic study was then conducted by liquid chromatography coupled with tandem mass spectrometry (LC/MS/MS) to simultaneously monitor 14 triterpenoids (**1**–**3**, **5**–**10**, **14**–**17**, **19**) and 2 metabolites (**M5**, **M6**). A brief workflow of this study is depicted in Figure 1.

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2. Materials and methods

2.1. Chemicals and reagents

Acetonitrile, methanol, and formic acid (Mallinkrodt Baker, Phillipsburg, NJ, USA) were of HPLC grade. De-ionized water was obtained from a Milli-Q system (Millipore, MA, USA). High-purity nitrogen (99.9%) and helium (99.99%) were purchased from Gas Supplies Center of Peking University Health Science Center (Beijing, China). Reference compounds **1**–**10**, **13**–**17**, **19** and **20** were isolated from *Antrodia cinnamomea* by the authors [10]. Reference compounds **11**, **12**, **M5** and **M6** were synthesized in this study. Their structures are given in Figure 2. The internal standard (IS) ganoderic acid B was purchased from Zelang Co. Ltd. (Nanjing, China).

2.2. Fungal materials and drug samples for animal treatment

The fruiting bodies of *Antrodia cinnamomea* were cultivated by Professor Yew-Min Tzeng at Chaoyang University of Technology, Taiwan. A voucher specimen (YMT 1002) was deposited at the Herbarium of School of Pharmaceutical Sciences, Peking University, Beijing, China. For drug sample preparation, 40 g of *A. cinnamomea* fruiting bodies were powdered and extracted with ethanol (800 mL × 2 hr × 3 times). The extract was evaporated to dryness *in vacuum*, suspended in de-ionized water (300 mL), and then extracted with ethyl acetate (300 mL × 3 times). The organic layer was evaporated to dryness *in vacuum* to produce 10.24 g of *A. cinnamomea* extract (ACE, 25.6% yield). Before oral administration to rats, ACE was suspended in 20 mL of 1% carboxymethylcellulose sodium salt solution (0.5 g/mL). The contents of triterpenoids

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in ACE were determined as described in Supplementary data.

*2.3. Semi-synthesis of metabolites **M5**, **M6**, **11** and **12***

A mixture of 25*S*-antcin B (**14**) and 25*R*-antcin B (**15**) (100 mg, 0.214 mmol) was reacted with $NaBH_4$ (11.5 mg, 0.321 mmol). The mixture was purified by semi-preparative HPLC with CH_3OH-H_2O-TFA gradient elution to obtain fraction SF-B. SF-B was subjected to semi-preparative HPLC (CH_3CN-H_2O-TFA, 0-100 min, 40-55% CH_3CN) to yield **M5** (5.24 mg, t_R=70.0 min), **M6** (2.33 mg, t_R=72.8 min), **11** (5.86 mg, t_R = 88.0 min), and **12** (4.03 mg, t_R = 89.5 min).

NMR spectra (400 MHz for ^{1}H and 100 MHz for ^{13}C) were obtained on a Bruker 400 MHz spectrometer in pyridine-d_5 with TMS as reference. NOE experiments were conducted on an Inova 600 MHz spectrometer. HRESIMS data were acquired on a Bruker APEX IV FT-MS spectrometer. Semi-preparative HPLC was performed on an Agilent 1200 instrument with a YMC Pack ODS-A column (250 mm × 10 mm, i.d., 5 μm, YMC Co. Ltd., Japan).

(25*S*)-3β-hydroxyl-7,11-di-*oxo*-4α-methylergost-8(9),24(28)-dien-26-oic acid (**M5**): white powder (MeOH); ^{1}H and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 469.2947 $[M-H]^-$ (calcd for $C_{29}H_{41}O_5$, 469.2949).

(25*R*)-3β-hydroxyl-7,11-di-*oxo*-4α-methylergost-8(9),24(28)-dien-26-oic acid (**M6**):

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white powder (MeOH); ^{1}H and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 469.2964 $[M-H]^-$ (calcd for $C_{29}H_{41}O_5$, 469.2949).

(25*R*)-antcin I (**11**): white powder (MeOH); ^{1}H and ^{13}C NMR data, see Table 1.

(25*S*)-antcin I (**12**): white powder (MeOH); ^{1}H and ^{13}C NMR data, see Table 1.

2.4. Animals

Male Sprague-Dawley rats (250 ± 10 g) were provided by the Experimental Animal Center of Peking University Health Science Center (Beijing, China). The rats were housed in a cage (465 × 300 × 200 mm) in a breeding room at 25°C, 60 ± 5% humidity, and a 12-hr dark-light cycle for 3 days, and were given free access to water and normal chow *ad libitum*. All animals were fasted for 24 hr before treatment. The animal facilities and protocols were approved by the Animal Care and Use Committee of Peking University Health Science Center. All procedures were in accordance with the National Academy of Sciences *Guide for the Care and Use of Laboratory Animals* [17].

2.5. Drug administration and sample treatment for metabolites identification

Antrodia cinnamomea extract (ACE) was orally administrated to rats (equivalent to 6.0 g/kg of the crude drug), and blood samples were collected from *arteria cervicalis* at 0.5, 1.5 and 12 hr after administration (*n*=2 for each time point). Single triterpenoid

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compounds, including antcin K (a mixture of **1** and **2**, 50 mg/kg for each of the epimer), (25*R*)-antcin C (**6**, 50 mg/kg), antcin H (a mixture of **7** and **8**, 50 mg/kg for each of the epimer), and antcin B (a mixture of **14** and **15**, 50 mg/kg for each of the epimer) were orally administrated to rats, respectively, and blood samples were collected from *arteria cervicalis* after 1.5 hr. Normal saline (2 mL) was given to rats as vehicle control, and blood were collected after 1.5 hr. The blood samples were immediately centrifuged at 6000 rpm (4°C) for 20 min to obtain plasma. For each sample, 2 mL of plasma was mixed with 6 mL of methanol, and centrifuged at 9000 rpm for 20 min. The supernatant was transferred to a clean test tube and dried under a gentle flow of nitrogen gas at 40°C. The residue was re-constituted in 200 μL of methanol. The samples were filtered through 0.22 μm nylon membranes. An 1-μL aliquot was injected for UHPLC/qTOF-MS analysis.

2.6. Drug administration for PK study of A. cinnamomea

For pharmacokinetic study, the rats (*n*=10) received a single oral dose of 1.0 g/kg of ACE (equivalent to 4.0 g/kg of the crude drug). Retro-orbital blood samples (about 350 μL) were collected into heparinized tubes at 0.17, 0.33, 0.5, 1, 2, 3, 6, 9, 12, 18, 24 and 36 hr after administration, and were immediately centrifuged at 6000 rpm (4°C) for 20 min to obtain plasma. The plasma samples were stored at -80°C before use. Samples were prepared following the procedures described in Section 2.8.

2.7. Calibration standards, quality control, and internal standard stock solutions

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Reference standards were dissolved in methanol to prepare individual stock solutions (1.00 mg/mL for **3**, **19** and **M6**; 2.00 mg/mL for the other compounds). These stock solutions were mixed and then serially diluted to obtain calibration standard (CS) stock solutions (100, 50, 25, 10, 5, 2.5, 1, 0.5, 0.25, 0.1, 0.05, 0.025, 0.01, and 0.005 μg/mL for each compound). Quality control (QC) stock solutions were prepared at three concentration levels as high-level QC (HQC), middle-level QC (MQC), and low-level QC (LQC), based on linear ranges of the analytes and the predicted concentrations in majority of samples. All LQC was set within 3 times of the lower limit of quantitation. Ganoderic acid B was used as the internal standard (IS), and was dissolved in acetonitrile to produce a 0.75 μg/mL IS stock solution. All solutions were sealed and stored at -20°C until use.

2.8. Plasma sample treatment

For CS and QC samples, 150 μL of CS or QC stock solution was added to 150 μL of blank rat plasma, followed by the addition of 150 μL of IS solution and 150 μL of methanol. For the other samples, 150 μL of plasma was mixed with 150 μL of IS solution and 300 μL of methanol. The above mixed solutions were vortexed (2200 rpm) for 1 min, ultrasonicated in a water bath for 5 min, and then centrifuged (9000 rpm, 4°C) for 10 min. The supernatant was separated and evaporated to dryness at 35°C using a speedvac concentrator. The residue was stored at -20°C, and reconstituted in 150 μL of acidic methanol (containing 6% of formic acid) before analysis to obtain the sample solution. After being filtered through 0.22 μm nylon

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membranes, a 5-μL aliquot was injected for LC/MS/MS analysis.

2.9. UHPLC/qTOF-MS analysis

High-accuracy mass spectra for metabolites identification were obtained on an Agilent 6540 qTOF mass spectrometer (Agilent Technologies, Waldbronn, Germany) connected to an Agilent series 1290 UHPLC instrument via an electrospray ionization (ESI) interface. The UHPLC instrument was equipped with a binary pump, a diode-array detector, an autosampler, and a column compartment. Samples were separated on an Agilent Zorbax Eclipse Plus C_{18} column (1.8 μm, 2.1 × 150 mm) with an online filter. The mobile phase consisted of acetonitrile (A) and water containing 0.1% formic acid (B). A linear gradient elution program was used as follows: 0 min, 30% A; 3 min, 58% A; 10 min, 58% A; 12 min, 70% A; 15 min, 95% A; 18 min, 95% A. The column temperature was 50°C. The flow rate was 0.3 mL/min. The ESI source was operated in the negative ion mode. High-purity nitrogen (N_2) was used as both drying gas and nebulizing gas, and ultra-high purity helium (He) was used as the collision gas. Other parameters were as follows: capillary voltage, 4000 V; nozzle voltage, 300 V; fragmentor voltage, 135 V; skimmer voltage, 65 V; octopole 1 rf voltage, 750 V; data acquisition, 3 spectra/s.

2.10. LC/MS/MS analysis for pharmacokinetic study

The LC/MS/MS system consisted of a Finnigan Surveyor LC instrument connected to a Finnigan TSQ Quantum triple quadrupole mass spectrometer via an ESI interface

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(ThermoFisher, CA, USA). The mobile phase consisted of acetonitrile (A) and water containing 0.1% formic acid (B). Samples were separated on a Symmetry C_{18} column (3.5 µm, 2.1 × 150 mm) (Waters, MA, USA) equipped with an Agilent Zorbax Extend-C_{18} guard column (5 µm, 2.1 × 12.5 mm). The following gradient elution program was used: 0-6 min, 38% A; 6-6.5 min, 38-52% A; 6.5-30 min, 52% A; 30-33 min, 52-100% A; 33-48 min, 100% A. The flow rate was 200 µL/min. The effluent was introduced into the mass spectrometer without splitting. The column temperature was set at 20°C. The mass spectrometer was operated in the negative ion mode. High purity nitrogen was used as the sheath (50 arb) and auxiliary (10 arb) gas, and high purity argon was used as the collision gas (1.5 mTorr). Other parameters were as follows: spray voltage, 3.1 kV; capillary temperature, 320°C; tube lens offset, -35 V. Q1 and Q3 quadrupoles were set at unit resolution. The analytes were detected by using the selected reaction monitoring (SRM) scan mode. The SRM parameters are listed in Table 2.

2.11. Data analysis

Calibration and quantitation data were processed by Xcalibur 2.0.7 software (ThermoFisher, CA, USA). Peak areas were obtained by using the automatic integration function ("toggle peak detection") provided by the software. High-resolution mass spectrometry data were analyzed by MassHunter software (Agilent Technologies). Mean resident time from 0 hr to the last detectable point (MRT_{last}) was calculated by non-compartment modeling using WinNonlin® software

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(v6.1, Pharsight, CA, USA). The maximal plasma concentrations (C_{max}) and their time of occurrence (T_{max}) were obtained directly from the measured data. The area under the plasma concentration-time curve (AUC_{last}) was calculated according to the linear trapezoidal rule to the last measurable concentration.

3. Results

3.1. Metabolites identification for Antrodia cinnamomea in rats

After oral administration of 1.5 g/kg *Antrodia cinnamomea* extract (equivalent to 6.0 g/kg of crude drug), the rats plasma samples were analyzed by UHPLC/qTOF-MS. The total ion current chromatogram for a 1.5-hr plasma sample is shown in Figure 3, as it contained more metabolites than the 0.5-hr or 12-hr samples (Supplemental Figure 1). A total of 26 triterpenoids could be detected. All the 20 major triterpenoids in *A. cinnamomea* were present in the plasma, except for the two Δ^8 lanostanes **13** and **20** (compound **9** was detected only in 12-hr samples, and was not shown in Figure 3). Besides, 8 biotransformed metabolites were detected. Their structures were tentatively characterized by analyzing their UV and MS spectra (Table 3). Particularly, some triterpenoids with similar molecular weights could be differentiated by high-resolution mass spectrometry (HRMS), like compounds **5** ($C_{29}H_{42}O_5$, $[M-H]^-$ *m/z* 469.2969), **9** ($C_{30}H_{46}O_4$, *m/z* 469.3328), and **20** ($C_{31}H_{50}O_3$, *m/z* 469.3696). To further support the characterization of metabolites, we obtained **M5**, **M6**, **11** and **12** by semi-synthesis, and unambiguously identified their structures by NMR spectroscopic analysis. We also studied the metabolism of single triterpenoids (antcins B, C, H and K), and thus

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assigned the parent compounds for **M1**–**M6**. Since that ergostanes are present as 25*R*/*S* epimeric pairs in *A. cinnamomea*, and were difficult to be purified, **1**/**2**, **7**/**8** and **14**/**15** were administered to rats for metabolites identification as epimeric mixtures.

HRMS established the molecular formula of **M5**, **M6**, **11** and **12** as $C_{29}H_{42}O_5$, two more hydrogen atoms than antcin B (**14/15**, $C_{29}H_{40}O_5$). The MS/MS spectra for their $[M-H]^-$ ions (*m*/*z* 469) showed two major fragments at *m/z* 299 and 409, which were 2 Da heavier than the corresponding fragments of antcin B (*m/z* 297 and 407, Supplemental Figure 2). The UV spectra showed maximum absorption at 270–272 nm, indicating the presence of an 8(9)-ene-7,11-dione substructure. **M5**, **M6**, **11** and **12** could be detected in rats plasma after oral administration of antcin B, indicating they were C-3 hydrogenated products of antcin B (Supplemental Figure 3). These four compounds were semi-synthesized by reduction of 25*R*/*S* antcin B with $NaBH_4$. In their ^{13}C NMR spectra, the carbonyl resonance corresponding to C-3 (δ_C 209.9) in antcin B disappeared, while an oxygen-bearing methine resonance appeared at δ_C 74.7 (**M5**/**M6**) or δ_C 69.2 (**11**/**12**). These data supported the presence of a 3-OH group. For **M5**/**M6**, H-3 (δ_H 3.24) showed a broad peak with half-height width of 32 Hz due to axial-axial couplings of H-3a with H-2a and H-4a, indicating a β-configuration of 3-OH [18]. For **11**/**12**, however, H-3 (δ_H 3.88) showed a narrow peak with half-height width of 4 Hz, indicating an α-configuration of 3-OH (Figure 4). **M5** and **M6**, as well as **11** and **12**, showed almost identical NMR spectra, indicating they were C-25 epimers. To determine the absolute configuration of C-25, 25*S*-antcin B (**14**, 3 mg,

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0.006 mmol) and 25*R*-antcin B (**15**, 3 mg, 0.006 mmol) were reduced by $NaBH_4$ (0.35 mg, 0.01 mmol), respectively, and the reaction mixtures were analyzed by LC/MS. **M5** and **12** were detected as reduction products of **14**, whereas **M6** and **11** were products of **15** (Figure 4). Thus, **M5** and **M6** were respectively identified as (25*S*)- and (25*R*)-3β-hydroxyl antcin B, which were reported for the first time. Compounds **11** and **12** were respectively identified as (25*R*)- and (25*S*)- antcin I (3α-hydroxyl antcin B), which had been isolated from *Antrodia cinnamomea* [19]. They were obtained in pure optical form for the first time.

M1, **M2** and **M4** were detected in rats plasma samples after oral administration of antcin B. According to their HRMS spectral data, they should be reduction and/or hydroxylation metabolites of antcin B. Here we take **M4** as example to describe their structural characterization. **M4** was a hydroxylated metabolite of antcin B. It showed maximum UV absorption at 274 nm, indicating the presence of an 8(9)-ene-7,11-dione substructure. Its MS/MS spectra showed major product ions at *m*/*z* 247 (ring C cleavage), *m*/*z* 295 (ring D cleavage), and *m*/*z* 83 (ring A cleavage). These fragments indicated that rings A, C and D of **M4** were similar to **3**, and that the hydroxyl group may be substituted at ring B (like C-6) (Supplemental Figure 2). Due to its complicated tandem mass spectrometry fragmentations, the position of substitution for **M1**, **M2** and **M4** could not be fully established. Most *Antrodia* ergostanes were present as 25*R*/*S* epimers. However, only one peak was detected for **M1**, **M2** and **M4** in this study. This may be due to poor chromatographic resolution of

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their 25*R*/*S* forms.

Likewise, **M3** was characterized as a monohydroxylated metabolite of antcin H (Supplemental Figure 2). **M7** and **M8** ($[M-H]^-$ *m/z* 455.3172 and 455.3174, calcd $C_{29}H_{44}O_4$) contained two more hydrogens than antcin A (**17**, $C_{29}H_{42}O_4$). They could be derived from antcin A, though its metabolism was not studied in this work due to limited amount. Based on the metabolic pathway of antcin B, the structures for **M7** and **M8** were proposed to be 3-hydroxyl antcin A.

Compound **18** ($C_{33}H_{52}O_5$, $[M-H]^-$ *m/z* 527.3754) contains two more hydrogens than **17**. Both compounds gave the same base peak at *m/z* 59 in the MS/MS spectra, indicating the presence of an acetoxyl group. The ring-cleaved fragments of **18** (*m/z* 397, 467) were also 2 Da heavier than the corresponding product ions of **17** (*m/z* 395, 465). Thus, **18** was tentatively characterized as the known versisponic acid D [20], which contains an 8(9)-ene (Δ^8) substructure instead of 7(8),9(11)-diene ($\Delta^{7,9}$) for **17**.

In total, 18 triterpenoids of *A. cinnamomea* and 8 metabolites were detected in rats plasma after oral administration of *A. cinnamomea* extract. Hydrogenation (reduction of carbonyl groups) and hydroxylation were the major metabolic reactions for ergostanes. The lanostanes showed relatively poor bioavailabilities. The $\Delta^{7,9}$ lanostanes (**9**, **10**, **17**, **19**) were found at low amounts in their unchanged form, while Δ^8 lanostanes (**13**, **20**) were not detected in the plasma samples. We also searched for

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phase II metabolites of the triterpenoids in plasma, urine and feces samples by LC/MS ($[M-H+80]^-$ and $[M-H+176]^-$, see Supplemental Figure 4). No phase II metabolites were detected.

3.2. Pharmacokinetics of Antrodia cinnamomea

The pharmacokinetics of 16 plasma-exposed triterpenoids of *A. cinnamomea* were studied. These compounds included 10 ergostanes (**1**–**3**, **5**–**8**, **14**–**16**), 4 $\Delta^{7,9(11)}$ lanostanes (**9**, **10**, **17**, **19**), and 2 ergostane metabolites (**M5**, **M6**). An LC/MS/MS method was established to simultaneously monitor these 16 compounds by using the selected reaction monitoring (SRM) scan mode. A typical chromatogram is shown in Supplemental Figure 5. The method was validated according to the U.S. FDA guidance on bioanalytical method validation [21]. The calibration curves covered wide dynamic ranges (100 to 1667-fold) with good linearity ($r^2 > 0.98$). Considering the high concentrations of **1** and **2**, quadratic calibration curve models were used. The other analytes were calibrated by linear models (Supplemental Table 1). Accuracy and precision was assessed by sample analysis at three concentration levels in the same day ($n = 5$) and on three consecutive days. The measured concentrations were compared to the nominal concentrations to indicate the accuracy, and the RSD of parallel analyses were calculated to indicate the precision. As shown in Supplemental Table 2, the accuracy ranged from 85.1-110.5% for LQC, MQC, and HQC, except that the accuracy of **1** in HQC was 81.1%. For compounds **3**, **7**, **8**, **M5** and **M6**, a wide dynamic range was required due to their higher plasma concentration than the HQC

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sample. Separation of 25*S*/*R* ergostane epimers (**1**/**2**, **5**/**6**, **8**/**7** and **14**/**15**) are challenging for RP-HPLC since each pair of epimers have very similar chromatographic behaviors. Though their peak areas could be obtained by auto-integration, incomplete separation might compromise their dynamic range and accuracy, especially when the concentration is high. RSD values for the analytes were no higher than 17.9% for LQC, and no higher than 14.5% for HQC and MQC, indicating acceptable variation of the method. Matrix effects and extraction efficiency were examined by three groups of standard addition experiments. For each group, three concentrations were added. In group 1, analytes were added to plasma and then extracted as QC samples; in group 2, analytes were added into post-extracted blank rat plasma matrix; in group 3, analytes were dissolved in methanol. Extraction efficiency was calculated as: EE (%) = (Peak area of group 1) / (Peak area of group 2) × 100%. The matrix effect was calculated as: ME (%) = (Peak area of group 2) / (Peak area of group 3) × 100%. Extraction efficiency of the 16 analytes varied within ± 20% at all three concentrations. Matrix effects were not significant for most analytes, with ion suppression ranged from -28.5 to 22.8% (Supplemental Table 3). Stability was determined at high and low concentrations after 24 hr storage in the sample tray, and after 10-day storage at -80ºC. All analytes were proved stable during analysis (variations between -16.0 and 15.9%, see Supplemental Table 4).

By using the validated LC/MS/MS method, ergostanes **1**–**3**, **5**–**8**, **14**–**16**, lanostanes **9**, **10**, **17**, **19**, and metabolites **M5** and **M6**, were monitored in rats plasma after oral

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administration of ACE (equivalent to 4.0 g/kg crude drug). Semi-logarithmic time-concentration plots are shown in Figure 5. We calculated the pharmacokinetic parameters by WinNonlin software using non-compartment models, as shown in Table 4. Antcin A (**16**) showed fairly low concentrations, and could be detected in plasma samples of only a few time points. Several points of antcin B (**14**, **15**) were below the lowest dynamic range. Therefore, some of their pharmacokinetic parameters were not calculated.

4. Discussion

Although *Antrodia cinnamomea* is widely used as adjuvant therapeutic agent and dietary supplement, its metabolism has rarely been revealed. Only the plasma distribution of two maleic anhydride derivatives (antrodin B and C) had been reported [22]. Triterpenoids accounted for around 10% of the drug materials, and should be its major bioactive constituents. In this work, we used UHPLC/qTOF-MS and LC/MS/MS to elucidate the metabolism and pharmacokinetics of *A. cinnamomea* in rats. Due to its complicated chemical composition, the metabolites identification was challenging, particularly when no reference standards were available. For reduction and hydroxylation of the triterpenoids, the position and stereo-configuration of the newly introduced hydroxyl groups were difficult to be identified by mass spectrometry. Therefore, we obtained two major metabolites **M5** and **M6** by $NaBH_4$ reduction of 25*R*/*S* antcin B, and unambiguously identified their absolute configuration. We also used our “From single compounds to herbal extract” strategy

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to facilitate structural characterization of unknown metabolites [23,24]. The metabolites of single compounds (antcins B, C, H, and K) were characterized, and the results could not only help identify the structures of unknown metabolites, but also assist in establishing the metabolic pathways of ergostanes (Figure 1).

The proposed metabolic pathways for antcins B, C, H and K are depicted in Figure 6. Antcin B mainly undertook hydrogenation and hydroxylation in rats. The carbonyl group at C-3 could be readily hydrogenated into 3β-OH to produce **M5**/**M6**, or 3α-OH to produce **11**/**12**. **M5**/**M6** were more abundant than **11**/**12** in plasma samples, probably due to higher stability of 3β-OH (equatorial bond) than 3α-OH (axial bond). The carbonyl group at C-7 could be hydrogenated into 7β-OH to produce antcin C (**5**/**6**). The carbonyl at C-11 of antcin B appeared to be difficult to be reduced. Hydrogenation of both carbonyl groups at C-3 and C-7 could produce **3**/**4**. Hydroxylation of antcin B and its hydrogenated metabolites produced **M1**, **M2** and **M4** (Supplemental Figure 3). The metabolic pathway for antcin C was very similar to that of antcin B. It could be reduced at C-3 to produce **3**/**4**, and could also undertake hydroxylation and reduction to yield **1**/**2**. Although antcin H contains a carbonyl group at C-7, it only produced one hydroxylated metabolite **M3**. This result indicated the carbonyl group at C-7 was more metabolically stable than the carbonyl at C-3. No metabolites were detected for antcin K (**1**/**2**), which only contains a carbonyl group at C-11.

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Since the hydrogenation of C-3 or C-7 carbonyl groups was a major metabolic reaction for the ergostanes, antcin B yielded more metabolites than antcins C or H. With no carbonyl group at C-3 or C-7, antcin K did not produce detectable metabolites. We also noticed that one ergostane could be metabolized into other ergostanes of *Antrodia cinnamomea*. For instance, antcamphins K/L (**3**/**4**) could be produced by both antcins B and C. This may be the reason why antcamphins K/L were detected in rats plasma at high abundances though they were present in *A. cinnamomea* in fairly low amounts. The same situation applies to antcin I (**11**/**12**), which could be derived from antcin B (Figure 3).

Given that no metabolites of lanostanes were detected after oral administration of *A. cinnamomea* extract, we deduced that lanostanes were metabolically stable. We fed the rats with compounds **10** and **19** at a dosage of 150 mg/kg, but did not detect any biotransformed metabolites, either. We also incubated compounds **10**, **17** and **19** in rat liver microsomes following the methods we have published [25], and only a very few portion of **17** was metabolized into hydroxylated products (Supplemental Figure 6). No metabolites for **10** and **19** were detected. The metabolism of lanostanes in *A. cinnamomea* was remarkably different from those in *Ganoderma lucidum* or *Poria cocos* [15,16].

The pharmacokinetic patterns of *Antrodia* triterpenoids were closely related with their structural features. It appeared that ergostanes with more hydroxyl groups and fewer

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carbonyl groups exhibited higher plasma exposure. As shown in Table 4, antcin K (**1**/**2**, 1 × C=O, 3 × OH) and antcin H (**7**/**8**, 2 × C=O, 2 × OH) showed remarkably higher C_{max} and *AUC* values than antcin B (**14**/**15**, 3 × C=O, no OH) and antcin C (**5**/**6**, 2 × C=O, 1 × OH). As hydrogenated products of antcins B and C, **M5**, **M6** and **3** also showed high maximum plasma concentrations (21.0-34.0 μM). All the ergostanes could be readily absorbed into circulation, with T_{max} of 0.5-1.0 hr. They were then rapidly eliminated, with *MRT* shorter than 8 hr, except for the metabolically stable antcin K (9-10 hr). Most ergostanes in *A. cinnamomea* occur as 25*R*/*S* epimers [26]. In this study, we found the epimers showed very similar pharmacokinetic curves, as exemplified by 25*R*/*S* forms of antcins B, C, H and K (Figure 5).

The PK patterns for lanostanes were remarkably different from those of ergostanes. Both the absorption and excretion were slow, with T_{max} of 2.1-10.9 hr and *MRT* of 9.2-16.2 hr. Although the C_{max} for **19** was only 3.1 μM, its *AUC* was similar to most ergostanes (64.3 hr*μM). The dose, *MRT*, and *AUC* of 15 ergostanes and lanostanes (except for antcin A) were summarized in Figure 7. Generally, ergostanes in *A. cinnamomea* showed much higher maximal plasma concentrations than lanostanes, and higher bioavailabilities than lanostanes.

The above results indicated that antcins K and H were the major exposure compounds of *A. cinnamomea* after oral administration. They had been reported to show anti-inflammatory and anti-cancer activities, and the IC_{50} values varied from 10 to 40

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μM [10,11,27]. According to our pharmacokinetic study, the maximal plasma concentrations of antcin K (81.0/91.3 μM for **1**/**2**) and antcin H (27.3/18.5 μM for **7**/**8**) were higher than or comparable to the IC_{50} values (Figure 7). They were also abundant in *A. cinnamomea* extract. Thus, antcins K and H may play a major role in the therapeutic effects of *A. cinnamomea.* Metabolites **M5**, **M6** and **3** also showed high plasma concentrations, and their bioactivities warrants future evaluation.

5. Conclusions

The metabolism and pharmacokinetics of *Antrodia cinnamomea* in rats was studied. A total of 26 triterpenoids and metabolites were detected in rats plasma after oral administration. The ergostanes and $\Delta^{7,9(11)}$ lanostanes, but not the Δ^{8} lanostanes, could get into circulation. Different types of triterpenoids showed remarkably different pharmacokinetic patterns. The ergostanes were generally absorbed and eliminated rapidly, whereas the lanostanes remained in the plasma at a low concentration for a relatively long time. High-polarity ergostanes (including antcins H and K) were the major plasma-exposed components of *A. cinnamomea*, and may play an important role in its therapeutic effects. The bioactivities of these compounds warrant further investigation to elucidate the mechanism of action of *A. cinnamomea*.

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Figure Captions

Fig. 1. A brief workflow of this study.

Fig. 2. Structures of reference compounds used in this study.

1, (25*S*)-antcin K; **2**, (25*R*)-antcin K; **3**, antcamphin K; **4**, antcamphin L; **5**, (25*S*)-antcin C; **6**, (25*R*)-antcin C; **7**, (25*R*)-antcin H; **8**, (25*S*)-antcin H; **9**, 3β,15α-dihydroxylanosta-7,9(11),24-triene-21-oic acid; **10**, dehydrosulphurenic acid; **11**, (25*R*)-antcin I; **12**, (25*S*)-antcin I; **13**, sulphurenic acid; **14**, (25*S*)-antcin B; **15**, (25*R*)-antcin B; **16**, antcin A; **17**, 15α-acetyldehydrosulphurenic acid; **19**, dehydroeburicoic acid; **20**, eburicoic acid; **M5**, (25*S*)-3β-hydroxyl-7,11-di-*oxo*-4α-methylergost-8(9),24(28)-dien-26-oic acid; **M6**, (25*R*)-3β-hydroxyl-7,11-di-*oxo*-4α-methylergost-8(9),24(28)-dien-26-oic acid; **IS**, internal standard, ganoderic acid B.

Fig. 3. UHPLC/qTOF-MS chromatograms of *Antrodia cinnamomea* extract (ACE), rat plasma sample after oral administration of ACE (1.5 hr), and extracted ion chromatograms (EIC) for major metabolites.

Ion extraction width was 10 mDa; ergostanes that showed increased relative abundance in plasma compared to ACE were indicated by red arrows; ergostanes that showed decreased relative abundance in plasma compared to ACE were indicated by green arrows.

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Fig. 4. Structural characterization of **M5**/**M6** and **11**/**12**. (A) ^{1}H NMR resonance for H-3; (B) ^{13}C NMR resonance for C-3; (C) HPLC/(-)-ESI-MS analysis of the $NaBH_4$ reduction product of 25*S*/*R*-antcin B (**14**/**15**).

LC conditions: Waters Symmetry C_{18} (4.6 × 250 mm, 5 μm), 0-45 min, 66-75% methanol in water containing 0.1% formic acid. The flow rate was 0.8 mL/min.

Fig. 5. Time-plasma concentration curves for 15 triterpenoids.

Y-axis refers to plasma concentration (μM) from 1×10^{-2} (1E-2) to 1×10^{3} (1E+3). MRT values of **14** and **15** were not calculated due to their low concentration.

Fig. 6. Proposed metabolic pathways for antcins B, C, H, and K.

Compounds tentatively characterized by mass spectrometry are shown in square brackets; major metabolic reactions were indicated with bold arrows; reduction and hydroxylation reactions were indicated in red solid arrows and blue dashed arrows, respectively.

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Note: Left *Y*-axis, dosage in absolute value; right *Y*-axis, AUC_{last} and MRT_{last} in relative value (to AUC_{last} of **2** and MRT_{last} of **19**, respectively). Dosages were determined by UPLC-UV.

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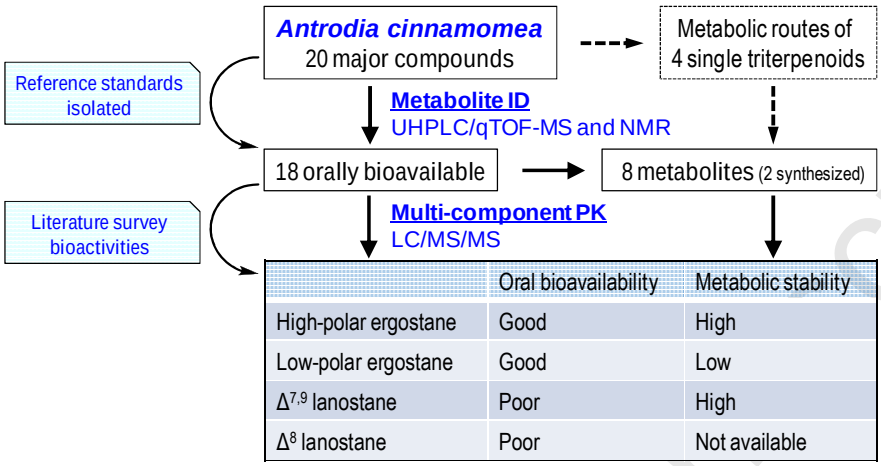
Figures in Enhanced Metafile (For Review Only)



	Oral bioavailability	Metabolic stability
High-polar ergostane	Good	High
Low-polar ergostane	Good	Low
$\Delta^{7,9}$ lanostane	Poor	High
Δ^{8} lanostane	Poor	Not available

Fig. 1. A brief workflow of this study.

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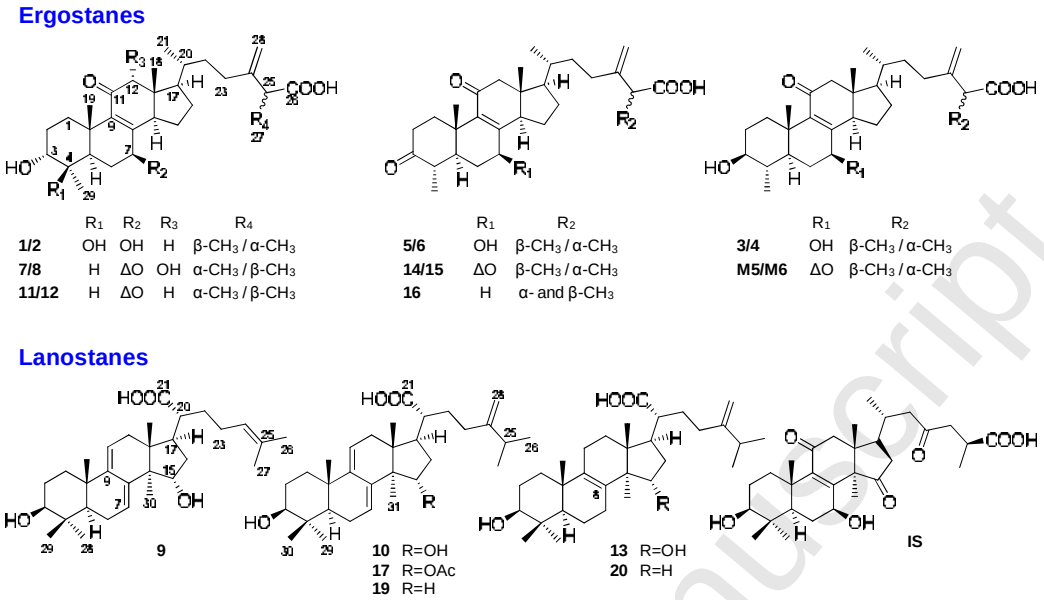

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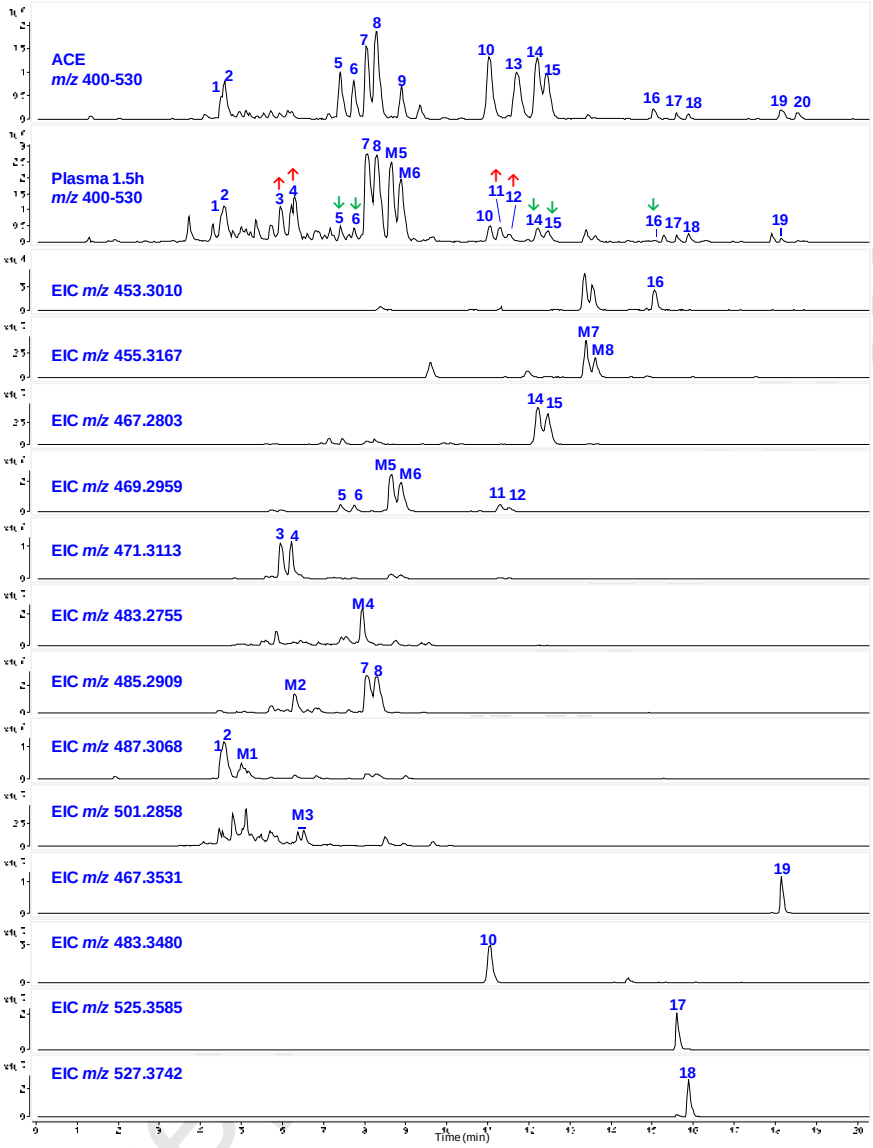

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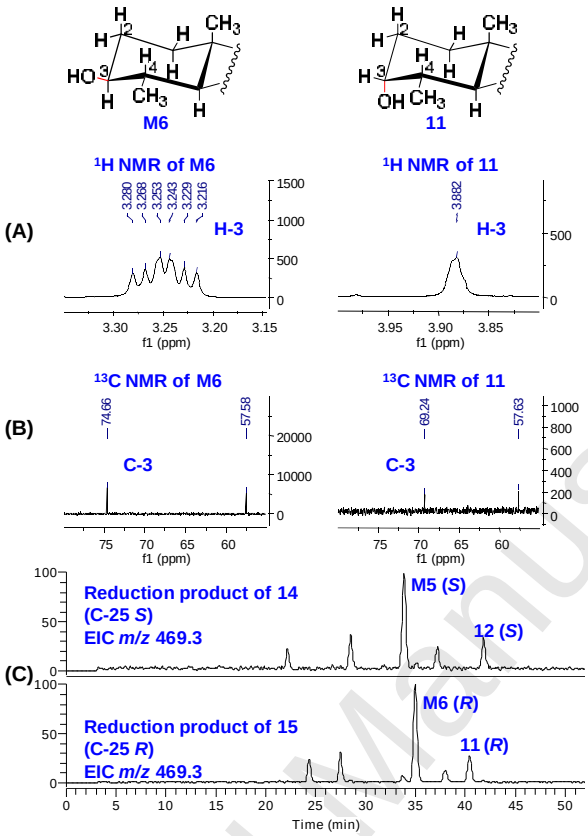

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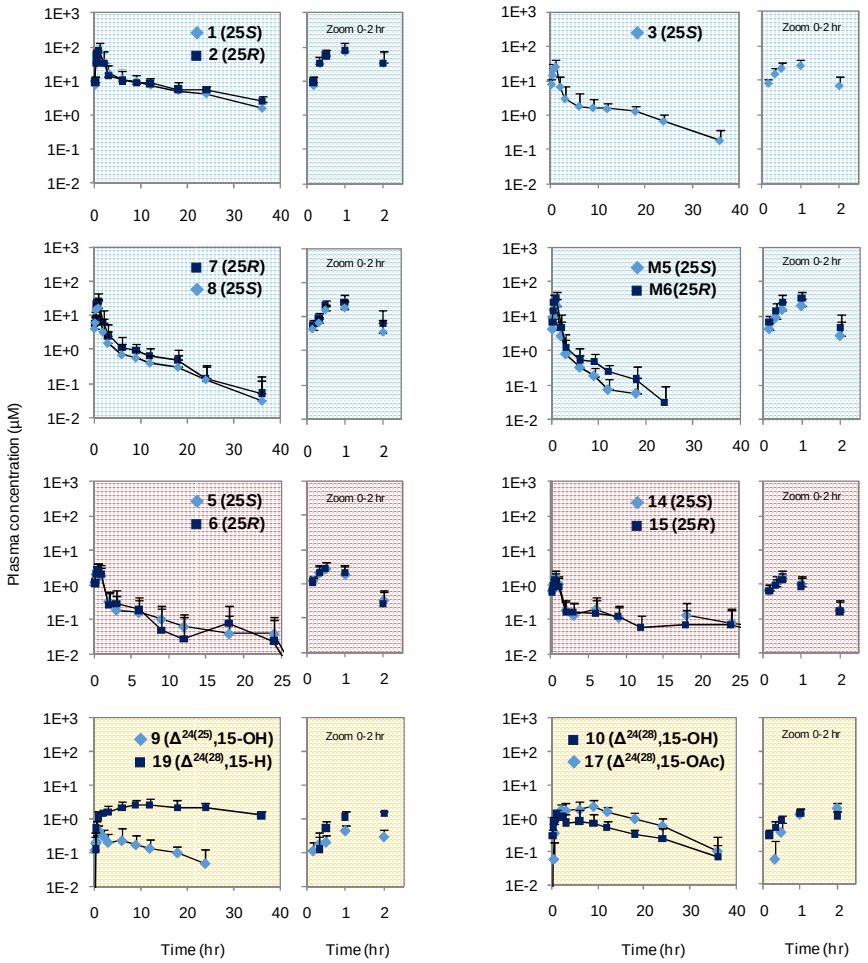

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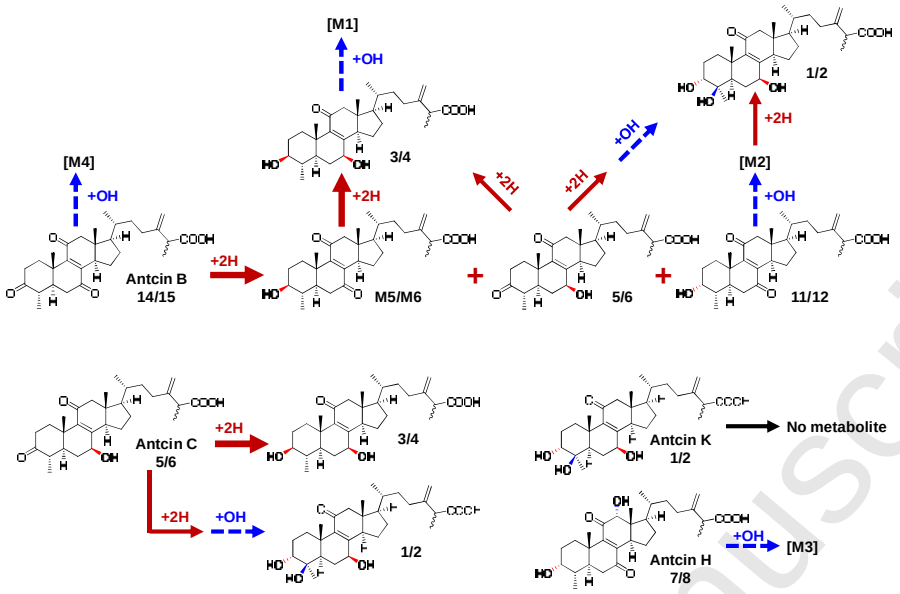

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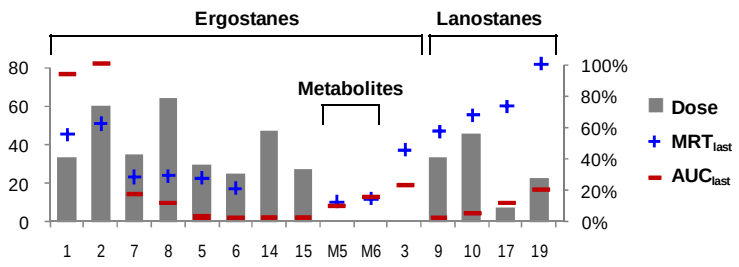

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Highlights

► We studied the DMPK of triterpenoids in *Antrodia cinnamomea*, an anticancer mushroom.

► In total 18 triterpenoids and 8 metabolites were identified by UHPLC/qTOF-MS.

► Two hydrogenated metabolites were obtained by semi-synthesis as new compounds.

► PK of 14 unchanged triterpenoids and 2 metabolites were studied by LC/MS/MS.

► High-polarity ergostanes are major plasma-exposed components of *A. cinnamomea*.

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Tables

Table 1. NMR spectroscopic data (400 MHz in pyridine-d_5) for antcin B (**14**/**15**) and their metabolites **11**/**12**, and **M5**/**M6**.

No.	**14/15**		**M5/M6**		**11/12**	
	δ_C	δ_H (J in Hz)	δ_C	δ_H (J in Hz)	δ_C	δ_H (J in Hz)
1	34.9 CH_2	3.15 m; 1.36 m	31.5 CH_2	2.48 m; 1.67 m	28.6 CH_2	2.69 m; 1.87 m
2	37.8 CH_2	2.54 m; 1.94 m	33.3 CH_2	1.82 m; 1.24 m	30.2 CH_2	1.84 m; 1.26 m
3	209.9 C		74.7 CH	3.24 m	69.2 CH	3.88 br s
4	43.9 CH	2.41 m	38.7 CH	1.66 m	35.3 CH	1.69 m
5	48.9 CH	1.85 m	47.3 CH	2.36 m	41.6 CH	2.58 m
6	39.2 CH_2	2.40 m; 2.36 m	38.8 CH_2	2.63 m; 2.42 m	38.5 CH_2	2.62 m; 2.41 m
7	200.8 C		201.9 C		202.3 C	
8	145.5 C		144.9 C		144.9 C	
9	151.9 C		153.4 C		153.9 C	
10	38.6 C		38.9 C		39.3 C	
11	202.7 C		202.7 C		202.9 C	
12	57.4 CH_2	2.99 d (14.0); 2.47 d (14.0)	57.6 CH_2	2.95 d (13.6); 2.48 d (13.6)	57.6 CH_2	2.96 d (13.6); 2.43 d (13.6)
13	47.2 C		47.5 C		47.4 C	
14	49.4 CH	2.72 m	49.5 CH	2.71 m	49.6 CH	2.72 m
15	25.3 CH_2	2.71 m; 1.50 m	25.3 CH_2	2.73 m; 1.51 m	25.4 CH_2	2.72 m; 1.52 m
16	28.0 CH_2	1.84 m; 1.21 m	27.9 CH_2	1.88 m; 1.21 m	28.0 CH_2	1.87 m; 1.23 m
17	53.9 CH_3	1.35 m	54.0 CH	1.37 m	54.1 CH	1.36 m
18	12.1 CH_3	0.67 s	12.0 CH_3	0.66 s	12.1 CH_3	0.68 s
19	16.2 CH_3	1.58 s	17.0 CH_3	1.42 s	16.2 CH_3	1.45 s
20	35.8 CH	1.36 m	35.8 CH	1.36 m	35.9 CH	1.36 m
21	18.5 CH_3	0.87 d (5.2)	18.5 CH_3	0.86 d (6.0)	18.5 CH_3	0.85 d (5.2)
22	34.2 CH_2	1.66 m; 1.19 m	34.3 CH_2	1.67 m; 1.21 m	34.3 CH_2	1.68 m; 1.22 m
23	31.8 CH_2	2.39 m; 2.19 m	31.7 CH_2	2.38 m; 2.17 m	31.6 CH_2	2.36 m; 2.17 m
24	150.5 C		150.3 C		150.4 C	
25	46.8 CH	3.45 q (7.2)	46.6 CH	3.43 q (6.8)	46.8 CH	3.45 q (6.8)
26	177.2 C		176.8 C		177.0 C	
27	17.1 CH_3	1.50 d (6.8)	17.1 CH_3	1.47 d (6.8)	17.1 CH_3	1.49 d (7.2)
28	110.3 CH_2	5.22 s; 5.06 s	110.5 CH_2	5.21 s; 5.04 s	110.4 CH_2	5.23 s; 5.05 s
29	11.9 CH_3	1.01 d (6.4)	15.2 CH_3	1.13 d (6.0)	16.5 CH_3	1.06 d (6.8)

Note: **14**/**15**, **12**/**11**, and **M5**/**M6** are 25*S*/*R* epimers, and their NMR spectra were identical.

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Table 2. Selected reaction monitoring (SRM) settings for 16 analytes and the internal standard (IS).

Analytes	TLO	Parent	Product_1	CE_1	Product_2	CE_2
1/2	168	487.2	443.3	29	407.2	36
3	191	471.2	427.3	33	409.3	39
5/6	191	469.2	425.3	33	247.2	44
7/8	150	485.3	413.3	37	423.3	39
9	189	469.2	469.2	15	269.1	41
10	194	483.2	483.2	15	269.1	46
14/15	148	467.2	407.2	46	423.2	31
16	145	453.1	409.2	34	393.2	47
17	192	525.3	465.2	43	401.2	49
19	174	467.2	467.2	15	337.2	38
IS	142	515.2	453.2	35	303.1	39
M5/M6	191	469.2	425.3	33	269.2	44

TLO, tube lens offset; CE, collision energy; source-induced dissociation energy was set at 10%; SRM transitions in Product_1 was used for quantitation; SRM transitions in Product_2 was used to confirm the identities of the analytes.

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Table 3. Characterization of metabolites in rats plasma after oral administration of *Antrodia cinnamomea* extract by UHPLC/qTOF-MS.

No.	t_R (min)	λ_{max} (nm)	Formula	$[M-H]^-$ Pred.	$[M-H]^-$ Meas.	Δ (ppm)	MS/MS of $[M-H]^-$	Identification	Parent compound [#]
1[*]	4.51	256	$C_{29}H_{44}O_6$	487.3068	487.3078	2.05	247, **259**, 273, 301, 407, 425, 443	(25*S*)-antcin K	antcin B/C
2[*]	4.65	256	$C_{29}H_{44}O_6$	487.3068	487.3079	2.26	**247**, 259, 273, 301, 407, 425, 443	(25*R*)-antcin K	antcin B/C
3[*]	5.93	254	$C_{29}H_{44}O_5$	471.3113	471.3116	0.64	83, 121, 247, **259**, 301, 409, 427	antcamphin K (25*S*)	antcin B/C
4[*]	6.21	254	$C_{29}H_{44}O_5$	471.3113	471.3123	2.12	83, 121, 247, **259**, 301, 409, 427	antcamphin L (25*R*)	antcin B/C
5[*]	7.43	256	$C_{29}H_{42}O_5$	469.2959	469.2969	2.13	**247**, 259, 407, 409, 425	(25*S*)-antcin C	antcin B
6[*]	7.76	256	$C_{29}H_{42}O_5$	469.2959	469.2972	2.77	**247**, 259, 407, 409, 425	(25*R*)-antcin C	antcin B
7[*]	8.08	270	$C_{29}H_{42}O_6$	485.2909	485.2915	1.24	297, 327, **383**, 413, 423	(25*R*)-antcin H	
8[*]	8.29	270	$C_{29}H_{42}O_6$	485.2909	485.2918	1.85	297, 327, **383**, 413, 423	(25*S*)-antcin H	
9[*]	8.91	244	$C_{30}H_{46}O_4$	469.3323	469.3328	1.07	**83**, 97, 269, 409	3β,15α-dihydroxylanosta-7,9(11),24-triene-21-oic acid	
10[*]	11.07	244	$C_{31}H_{48}O_4$	483.3480	483.3488	1.66	**83**, 97, 269, 353, 387	dehydrosulphurenic acid	
11[*]	11.26	272	$C_{29}H_{42}O_5$	469.2959	469.2968	1.92	201, 299, **409**, 425	(25*R*)-antcin I	antcin B
12[*]	11.54	272	$C_{29}H_{42}O_5$	469.2959	469.2970	2.34	201, 299, **409**, 425	(25*S*)-antcin I	antcin B
14[*]	12.23	270	$C_{29}H_{40}O_5$	467.2803	467.2809	1.28	271, 297, **407**, 423	(25*S*)-antcin B	
15[*]	12.48	270	$C_{29}H_{40}O_5$	467.2803	467.2811	1.71	271, 297, **407**, 423	(25*R*)-antcin B	
16[*]	15.08	256	$C_{29}H_{42}O_4$	453.3010	453.3018	1.76	271, 377, **393**, 409, 427	(25*S*/*R*)-antcin A	
17[*]	15.64	244	$C_{33}H_{50}O_5$	525.3585	525.3592	1.33	**59**, 395, 401, 465	15α-acetyldehydrosulphurenic acid	
18	15.87	w	$C_{33}H_{52}O_5$	527.3742	527.3754	2.28	**59**, 397, 467	versisponic acid D	
19[*]	18.14	244	$C_{31}H_{48}O_3$	467.3531	467.3540	1.93	323, **337**, 371, **467**	dehydroeburicoic acid	
M1	5.16	258	$C_{29}H_{44}O_6$	487.3068	487.3071	0.62	193, **235**, 259, 425, 443	antcin B +4H+OH	antcin B
M2	6.30	w	$C_{29}H_{42}O_6$	485.2909	485.2926	3.50	**149**, 257, 297, 339, 423, 441	antcin B +2H+OH	antcin B
M3	6.37	w	$C_{29}H_{42}O_7$	501.2858	501.2872	2.79	**83**, 95, 207, 247, 273, 381, 439	antcin H +OH	antcin H

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M4	7.99	274	$C_{29}H_{40}O_6$	483.2755	483.2764	1.86	**83**, 123, 247, 295, 383, 423	antcin B +OH	antcin B
M5[*]	8.64	270	$C_{29}H_{42}O_5$	469.2959	469.2967	1.70	269, 273, 299, **409**, 425	(25*S*)-antcin B +2H	antcin B
M6[*]	8.89	270	$C_{29}H_{42}O_5$	469.2959	469.2969	2.13	269, 273, 299, **409**, 425	(25*R*)-antcin B +2H	antcin B
M7	13.39	w	$C_{29}H_{44}O_4$	455.3167	455.3172	1.10	284, **395**, 411	antcin A +2H	
M8	13.64	w	$C_{29}H_{44}O_4$	455.3167	455.3174	1.54	284, **395**, 411	antcin A +2H	

[*], identified by comparing with reference standards.

[#], Parent compounds of the metabolites were identified by comparing with the metabolites of single compounds.

w, weak signals.

Base peaks of the MS/MS spectra were indicated in bold face.

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Table 4. Pharmacokinetic parameters for 16 triterpenoids of *Antrodia cinnamomea*.

	Dose (mg/kg)	C_{max} (μM)	T_{max} (hr)	AUC_{last} (hr*μM)	*MRT* (hr)
1	33.3	81.0 ± 40.8	1.0 ± 0.4	312.7 ± 91.2	8.9 ± 2.4
2	60.3	91.3 ± 55.4	1.0 ± 0.4	332.9 ± 102.8	10.0 ± 2.6
3	-	26.7 ± 15.2	0.9 ± 0.2	74.2 ± 28.9	7.3 ± 2.3
5	29.5	3.0 ± 1.9	0.7 ± 0.5	6.0 ± 3.4	4.4 ± 2.4
6	25.0	3.3 ± 1.9	0.6 ± 0.2	5.4 ± 3.2	3.3 ± 3.2
7	35	27.3 ± 15.9	0.9 ± 0.2	54.5 ± 22.8	4.5 ± 2.1
8	63.8	18.5 ± 12.0	0.9 ± 0.2	35.3 ± 15.8	4.6 ± 2.3
9	33.7	0.6 ± 0.2	2.1 ± 2.2	3.8 ± 1.9	9.2 ± 3.6
10	45.9	1.7 ± 0.6	3.2 ± 2.8	15.1 ± 5.2	10.9 ± 3.2
14	47.2	1.9 ± 1.1	0.6 ± 0.3	4.9 ± 1.6	Not available
15	27.4	1.5 ± 0.9	0.5 ± 0.3	4.1 ± 1.5	Not available
16	4.9	0.3 ± 0.1	0.7 ± 0.3	Not available	Not available
17	7.4	2.9 ± 0.8	6.9 ± 4.2	36.8 ± 7.0	11.8 ± 2.7
19	23.0	3.1 ± 1.3	10.9 ± 6.6	64.3 ± 28.0	16.2 ± 2.7
M5	-	21.0 ± 12.2	1.0 ± 0.2	29.1 ± 13.3	1.9 ± 0.7
M6	-	34.0 ± 19.9	1.0 ± 0.2	49.0 ± 22.6	2.2 ± 0.6

AUC_{last}, area under the plasma concentration–time curve from 0 hr to the last detectable point; MRT_{last}, mean resident time from 0 hr to the last detectable point.

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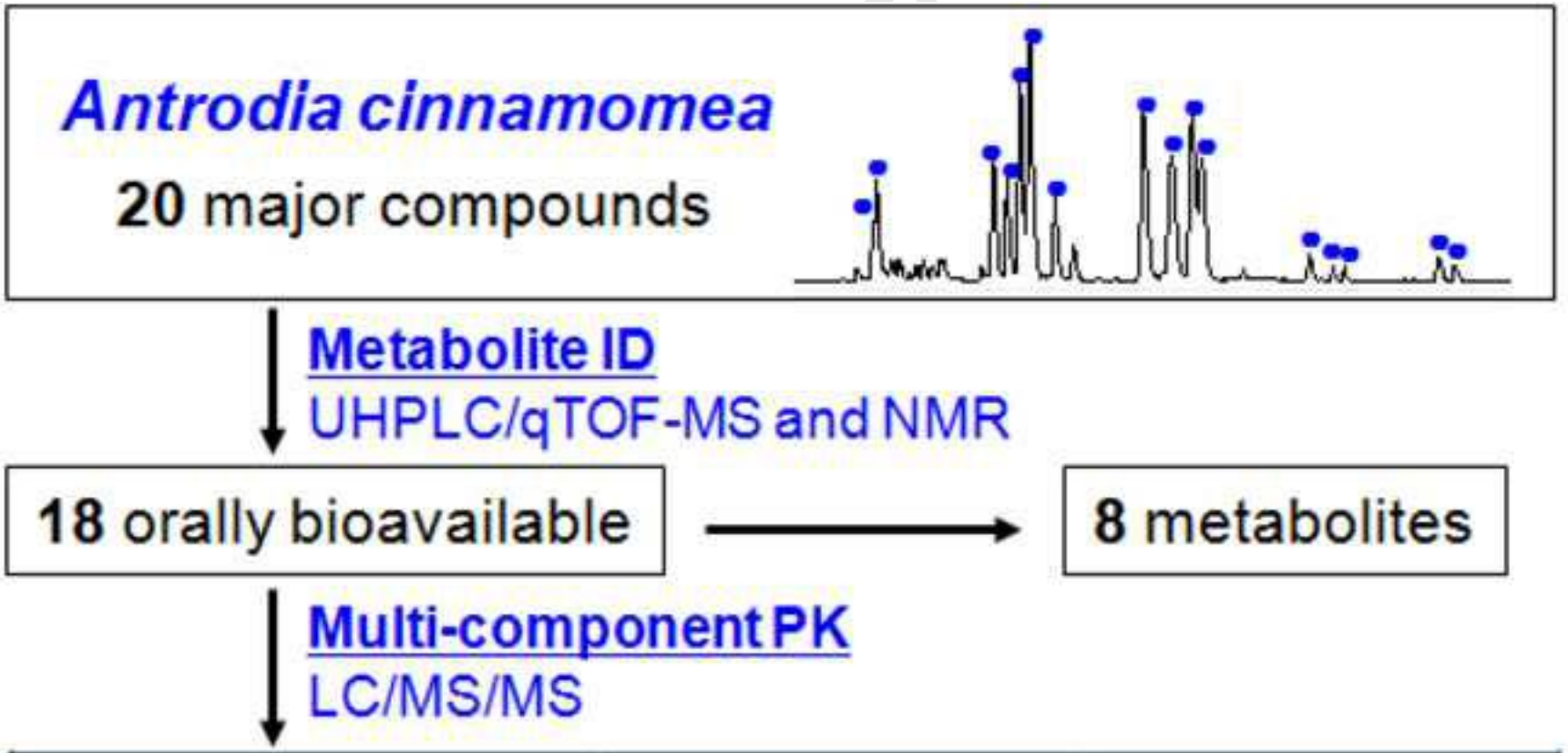
*Graphical Abstract



	Oral bioavailability	Metabolic stability
High-polar ergostane	Good	High
Low-polar ergostane	Good	Low
$\Delta^{7,9}$ lanostane	Poor	High
Δ^{8} lanostane	Poor	Not available